

Shiham Chowdhury

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EDUCATION

University of Michigan

Bachelor of Science in Electrical Engineering

– Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Databases, Embedded Systems

Ann Arbor, MI

Expected May 2027

TECHNICAL SKILLS

Languages: Java, C++, C#, Python, JavaScript, MATLAB

Developer Tools: Visual Studio, Jupyter Notebook, React, Git, LTSpice

Cloud/Databases: AWS IoT Core, AWS S3, AWS EC2, SQL (basic)

Soft: Verbal Communication Skills, Time Management, Critical Thinking

EXPERIENCE

eTitle

May 2025 – Aug 2025

Data Analyst Intern

Troy, MI

- Streamlined foreclosure management workflows by refactoring merge template scripts, reducing hard-coded dependencies and cutting template errors by 25%
- Partnered with senior developers in an Agile Scrum environment to test and validate 100+ foreclosure case templates, ensuring reliable document automation for clients
- Documented and presented process improvements to team, contributing to improved maintainability of the system

Wolverine Solar Car Team

Aug 2023 – Present

Battery Management Team

Ann Arbor, MI

- Prevented battery cells from exceeding 40 °C under high load by implementing a PID-controlled PWM fan system, doubling projected lifespan and reducing risk of thermal runaway
- Improved cooling system by designing a low-pass filter in MATLAB, reducing false fan activations by 30% and cutting temperature fluctuations by 20% for stable PID control
- Created unit tests and debug harnesses in C to validate control logic under edge cases, cutting false activations by 25% and improving reliability during endurance runs
- Analyzed battery cell voltage/temperature data using Python and pandas, identifying degradation patterns that informed battery replacement schedules
- Implemented linear regression models using scikit-learn to predict battery capacity fade based on charge cycles and temperature exposure from race data

PROJECTS

Environment Monitoring Device | *ESP32, AWS IoT Core, DynamoDB, S3, EC2, React*

- Programmed ESP32 microcontrollers with DHT22 sensors to send temperature/humidity readings via MQTT to AWS IoT Core
- Configured AWS IoT Rules Engine to route structured readings into DynamoDB for queries, while archiving raw DHT22 signal files in S3 for long-term storage
- Created preprocessing pipeline on EC2 using Python/MATLAB libraries to transform raw S3 signal files into graphs for frontend visualization
- Implemented FastAPI endpoints on EC2 with GET routes to fetch processed sensor data and metadata from DynamoDB for real-time visualization in the React dashboard
- Developed React dashboard with matplotlib and Chart.js to display live temperature/humidity readings in tables alongside signal visualizations from preprocessed S3 data

Room Occupancy Monitor | *ESP32, C++, IoT, I2C, UART*

- Designed a real-time, bidirectional occupancy monitor using ESP32 microcontrollers, dual E18-D80NK IR laser sensors and a VL53L5CX ToF sensor to reliably track people entering/exiting a room
- Developed a C++ finite-state machine that uses IR break order and ToF head detection to control count update logic, achieving robustness in complex scenarios such as u-turns or 20-person congo lines
- Implemented two-way wireless architecture to stream occupancy state to a server-hosting ESP32 over WiFi and support remote count resets via user serial input
- Optimized power consumption to allow system to run on battery for several days with minimal loss in performance
- Designed a compact, cost-optimized PCB to integrate all subsystems, enabling scalable production